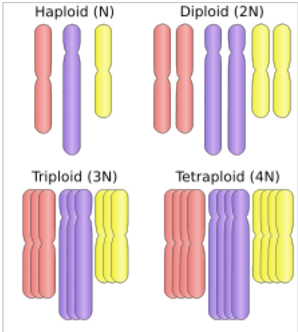
# Polyploid

From Wikipedia, the free encyclopedia




This image shows haploid (single), diploid (double), triploid (triple), and tetraploid (quadruple) sets of chromosomes. Triploid and tetraploid chromosomes are examples of polyploidy.

**Polyploid** cells and organisms are those containing more than two paired (homologous) sets of chromosomes. Most species whose cells have nuclei (Eukaryotes) are diploid, meaning they have two sets of chromosomes—one set inherited from each parent. However, **polyploidy** is found in some organisms and is especially common in plants. In addition, polyploidy occurs in some tissues of animals that are otherwise diploid, such as human muscle tissues.[1] This is known as **endopolyploidy**. Species whose cells do not have nuclei, that is, Prokaryotes, may be polyploid organisms, as seen in the large bacterium *Epulopicium fishelsoni* [1] (http://m.pnas.org/content/105/18/6730.full). Hence ploidy is defined with respect to a cell. Most eukaryotes have diploid somatic cells, but produce haploid gametes (eggs and sperm) by meiosis. A monoploid has only one set of chromosomes, and the term is usually only applied to cells or organisms that are normally diploid. Male bees and other Hymenoptera, for example, are monoploid. Unlike animals, plants and multicellular algae have life cycles with two alternating multicellular generations. The gametophyte generation is haploid, and produces gametes by mitosis, the sporophyte generation is diploid and produces spores by meiosis.

Polyploidy refers to a numerical change in a whole set of chromosomes. Organisms in which a particular chromosome, or chromosome segment, is under- or overrepresented are said to be aneuploid (from the Greek words meaning "not", "good", and "fold"). Therefore the distinction between aneuploidy and polyploidy is that aneuploidy refers to a numerical change in part of the chromosome set, whereas polyploidy refers to a numerical change in the whole set of chromosomes.[2]

Polyploidy may occur due to abnormal cell division, either during mitosis, or commonly during metaphase I in meiosis.

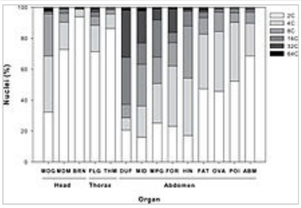
Polyploidy occurs in some animals, such as goldfish,[3] salmon, and salamanders, but is especially common among ferns and flowering plants (see *Hibiscus rosa-sinensis*), including both wild and cultivated species. Wheat, for example, after millennia of hybridization and modification by humans, has strains that are **diploid** (two sets of chromosomes), **tetraploid** (four sets of chromosomes) with the common name of durum or macaroni wheat, and **hexaploid** (six sets of chromosomes) with the common name of bread wheat. Many agriculturally important plants of the genus *Brassica* are also tetraploids.

Polyploidy can be induced in plants and cell cultures by some chemicals: the best known is colchicine, which can result in chromosome doubling, though its use may have other less obvious consequences as well. Oryzalin will also double the existing chromosome content.

## Contents



## Types




Organ-specific patterns of endopolyploidy (from $2x$ to $64x$) in the giant ant *Dinoponera australis*

**Polyploid** types are labeled according to the number of chromosome sets in the nucleus. The letter $x$ is used to represent the number of chromosomes in a single set.

- **triploid** (three sets; $3x$), for example seedless watermelons, common in the phylum Tardigrada[4]
- **tetraploid** (four sets; $4x$), for example Salmonidae fish,[5] the cotton *Gossypium hirsutum* [6]
- **pentaploid** (five sets; $5x$), for example Kenai Birch (*Betula papyrifera* var. *kenaica*)
- **hexaploid** (six sets; $6x$), for example wheat, kiwifruit[7]
- **heptaploid** or **septaploid** (seven sets; $7x$)
- **octaploid** or **octoploid**, (eight sets; $8x$), for example *Acipenser* (genus of sturgeon fish), dahlias
- **decaploid** (ten sets; $10x$), for example certain strawberries
- **dodecaploid** (twelve sets; $12x$), for example the plant *Celosia argentea* or the invasive one *Spartina anglica*[8] or the amphibian *Xenopus ruwenzoriensis*.

## Animals

Examples in animals are more common in non-vertebrates[9] such as flatworms, leeches, and brine shrimp. Within vertebrates, examples of stable polyploidy include the salmonids and many cyprinids (i.e. carp).[10] Some fish have as many as 400 chromosomes.[10] Polyploidy also occurs commonly in amphibians; for example the biomedically-important *Xenopus* genus contains many different species with as many as 12 sets of chromosomes (dodecaploid).[11] Polyploid lizards are also quite common, but are sterile and must reproduce by parthenogenesis. Polyploid mole salamanders (mostly triploids) are all female and reproduce by kleptogenesis,[12] "stealing" spermatophores from diploid males of related species to trigger egg development but not incorporating the males' DNA into the offspring. While mammalian liver cells are polyploid, rare instances of polyploid mammals are known, but most often result in prenatal death.

An octodontid rodent of Argentina's harsh desert regions, known as the Plains Viscacha-Rat (*Tympanoctomys barrerae*) has been reported as an exception to this 'rule'.[13] However, careful analysis using chromosome paints shows that there are only two copies of each chromosome in *T. barrerae*, not the four expected if it were truly a tetraploid.[14] The rodent is not a rat, but kin to guinea pigs and chinchillas. Its "new" diploid [2n] number is 102 and so its cells are roughly twice normal size. Its closest living relation is *Octomys mimax*, the Andean Viscacha-Rat of the same family, whose 2n = 56. It was therefore surmised that an *Octomys*-like ancestor produced tetraploid (i.e., 2n = 4x = 112) offspring that were, by virtue of their doubled chromosomes, reproductively isolated from their parents.

Polyploidy was induced in fish by Har Swarup (1956) using a cold-shock treatment of the eggs close to the time of fertilization, which produced triploid embryos that successfully matured.[15][16] Cold or heat shock has also been shown to result in unreduced amphibian gametes, though this occurs more commonly in eggs than in sperm.[17] John Gurdon (1958) transplanted intact nuclei from somatic cells to produce diploid eggs in the frog, *Xenopus* (an extension of the work of Briggs and King in 1952) that were able to develop to the tadpole stage.[18] The British Scientist, J. B. S. Haldane hailed the work for its potential medical applications and, in describing the results, became one of the first to use the word "clone" in reference to animals. Later work by Shinya Yamanaka showed how mature cells can be reprogrammed to become pluripotent, extending the possibilities to non-stem cells. Gurdon and Yamanaka were jointly awarded the Nobel Prize in 2012 for this work.[18]

### Humans

True polyploidy rarely occurs in humans, although polyploid cells occur in highly differentiated tissue, such as liver parenchyma and heart muscle, and in bone marrow.[19] Aneuploidy is more common.

Polyploidy occurs in humans in the form of triploidy, with 69 chromosomes (sometimes called 69,XXX), and tetraploidy with 92 chromosomes (sometimes called 92,XXXX). Triploidy, usually due to polyspermy, occurs in about 2–3% of all human pregnancies and ~15% of miscarriages. The vast majority of triploid conceptions end as a miscarriage; those that do survive to term typically die shortly after birth. In some cases, survival past birth may extend longer if there is mixoploidy with both a diploid and a triploid cell population present.

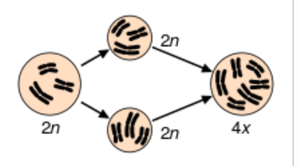
Triploidy may be the result of either digyny (the extra haploid set is from the mother) or diandry (the extra haploid set is from the father). Diandry is mostly caused by reduplication of the paternal haploid set from a single sperm, but may also be the consequence of dispermic (two sperm) fertilization of the egg.[20] Digyny is most commonly caused by either failure of one meiotic division during oogenesis leading to a diploid oocyte or failure to extrude one polar body from the oocyte. Diandry appears to predominate among early miscarriages, while digyny predominates among triploidy that survives into the fetal period. However, among early miscarriages, digyny is also more common in those cases <8.5 weeks gestational age or those in which an embryo is present. There are also two distinct phenotypes in triploid placentas and fetuses that are dependent on the origin of the extra haploid set. In digyny, there is typically an asymmetric poorly grown fetus, with marked adrenal hypoplasia and a very small placenta. In diandry, a partial hydatidiform mole develops.[20] These parent-of-origin effects reflect the effects of genomic imprinting.

Complete tetraploidy is more rarely diagnosed than triploidy, but is observed in 1–2% of early miscarriages. However, some tetraploid cells are commonly found in chromosome analysis at prenatal diagnosis and these are generally considered 'harmless'. It is not clear whether these tetraploid cells simply tend to arise during *in vitro* cell culture or whether they are also present in placental cells *in vivo*. There are, at any rate, very few clinical reports of fetuses/infants diagnosed with tetraploidy mosaicism.

Mixoploidy is quite commonly observed in human preimplantation embryos and includes haploid/diploid as well as diploid/tetraploid mixed cell populations. It is unknown whether these embryos fail to implant and are therefore rarely detected in ongoing pregnancies or if there is simply a selective process favoring the diploid cells.

## Plants

Polyploidy is pervasive in plants and some estimates suggest that 30–80% of living plant species are polyploid, and many lineages show evidence of ancient polyploidy (paleopolyploidy) in their genomes.[21][22][23] Huge explosions in angiosperm species diversity appear to have coincided with the timing of ancient genome duplications shared by many species.[24] It has been established that 15% of angiosperm and 31% of fern speciation events are accompanied by ploidy increase.[25]



Speciation via polyploidy: A diploid cell undergoes failed meiosis, producing diploid gametes, which self-fertilize to produce a tetraploid zygote.

Polyploid plants can arise spontaneously in nature by several mechanisms, including meiotic or mitotic failures, and fusion of unreduced (2n) gametes.[26] Both autopolyploids (e.g. potato [27]) and allopolyploids (e.g. canola, wheat, cotton) can be found among both wild and domesticated plant species.

Most polyploids display novel variation or morphologies relative to their parental species, that may contribute to the processes of speciation and eco-niche exploitation.[22][26] The mechanisms leading to novel variation in newly formed allopolyploids may include gene dosage effects (resulting from more numerous copies of genome content), the reunion of divergent gene regulatory hierarchies, chromosomal rearrangements, and epigenetic remodeling, all of which affect gene content and/or expression levels.[28][29][30][31] Many of these rapid changes may contribute to reproductive isolation and speciation. However seed generated from interploidy crosses, such as between polyploids and their parent species, usually suffer from aberrant endosperm development which impairs their viability,[32][33] thus contributing to polyploid speciation.

*Lomatia tasmanica* is an extremely rare Tasmanian shrub that is triploid and sterile; reproduction is entirely vegetative, with all plants having the same genetic constitution.

There are few naturally occurring polyploid conifers. One example is the Coast Redwood *Sequoia sempervirens*, which is a hexaploid (6x) with 66 chromosomes (2n = 6x = 66), although the origin is unclear.[34]

Aquatic plants, especially the Monocotyledons, include a large number of polyploids.[35]

### Crops

The induction of polyploidy is a common technique to overcome the sterility of a hybrid species during plant breeding. For example, Triticale is the hybrid of wheat (*Triticum turgidum*) and rye (*Secale cereale*). It combines sought-after characteristics of the parents, but the initial hybrids are sterile. After polyploidization, the hybrid becomes fertile and can thus be further propagated to become triticale.

In some situations, polyploid crops are preferred because they are sterile. For example, many seedless fruit varieties are seedless as a result of polyploidy. Such crops are propagated using asexual techniques, such as grafting.

Polyploidy in crop plants is most commonly induced by treating seeds with the chemical colchicine.

**Examples**

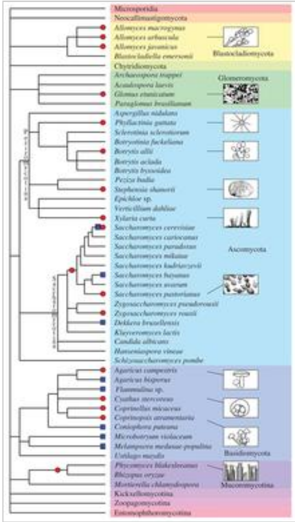
- Triploid crops: apple, banana, citrus, ginger, watermelon[36]
- Tetraploid crops: apple, durum or macaroni wheat, cotton, potato, canola/rapeseed, leek, tobacco, peanut, kinnow, Pelargonium
- Hexaploid crops: chrysanthemum, bread wheat, triticale, oat, kiwifruit[7]
- Octaploid crops: strawberry, dahlia, pansies, sugar cane, oca (*Oxalis tuberosa*)[37]
- Dodecaploid crops: some sugar cane hybrids [38]

Some crops are found in a variety of ploidies: tulips and lilies are commonly found as both diploid and triploid; daylilies (*Hemerocallis* cultivars) are available as either diploid or tetraploid; apples and kinnows can be diploid, triploid, or tetraploid.

# Fungi

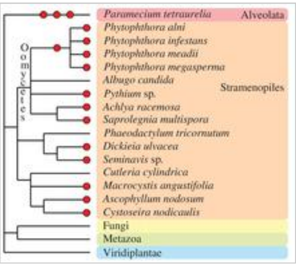
Besides plants and animals, the evolutionary history of various fungal species is dotted by past and recent whole-genome duplication events (see Albertin and Marullo 2012[39] for review). Several examples of polyploids are known:

- autopolyploid: the aquatic fungi of genus *Allomyces*,[40] some *Saccharomyces cerevisiae* strains used in bakery,[41] etc.
- allopolyploid: the widespread *Cyathus stercoreus*,[42] the allotetraploid lager yeast *Saccharomyces pastorianus*,[43] the allotriploid wine spoilage yeast *Dekkera bruxellensis*,[44] etc.
- paleopolyploid: the human pathogen *Rhizopus oryzae*,[45] the *Saccharomyces* genus,[46] etc.



Schematic phylogeny of the fungi. Red circles indicate polyploidy, blue squares indicate hybridization. From Albertin and Marullo, 2012[39]

In addition, polyploidy is frequently associated with hybridization and reticulate evolution that appear to be highly prevalent in several fungal taxa. Indeed, homoploid speciation (*i.e.*, hybrid speciation without a change in chromosome number) has been evidenced for some fungal species (*e.g.*, the basidiomycota *Microbotryum violaceum* [47]).




Schematic phylogeny of the Chromalveolata. Red circles indicate polyploidy, blue squares indicate hybridization. From Albertin and Marullo, 2012[39]

As for plants and animals, fungal hybrids and polyploids display structural and functional modifications compared to their progenitors and diploid counterparts. In particular, the structural and functional outcomes of polyploid *Saccharomyces* genomes strikingly reflect the evolutionary fate of plant polyploid ones. Large chromosomal rearrangements[48] leading to chimeric chromosomes[49] have been described, as well as more punctual genetic modifications such as gene loss.[50] The homoealleles of the allotetraploid yeast *S. pastorianus* show unequal contribution to the transcriptome.[51] Phenotypic diversification is also observed following polyploidization and/or hybridization in fungi,[52] producing the fuel for natural selection and subsequent adaptation and speciation.

# Chromalveolata

Other eukaryotic taxa have experienced one or more polyploidization events during their evolutionary history (see Albertin and Marullo, 2012[39] for review). The oomycetes, which are non-true fungi members, contain several examples of paleopolyploid and polyploid species, such as within the *Phytophthora* genus.[53] Some species of brown algae (Fucales, Laminariales [54] and diatoms [55]) contain apparent polyploid genomes. In the Alveolata group, the remarkable species *Paramecium tetraurelia* underwent three successive rounds of whole-genome duplication [56] and established itself as a major model for paleopolyploid studies.

# Terminology

## Autopolyploidy

Autopolyploids are polyploids with multiple chromosome sets derived from a single species. Autopolyploids can arise from a spontaneous, naturally occurring genome doubling, like the potato.[27] Others might form following fusion of 2n gametes (unreduced gametes). Bananas and apples can be found as autotriploids. Autopolyploid plants typically display polysomic inheritance, and therefore have low fertility, but may be propagated clonally.

## Allopolyploidy

Allopolyploids are polyploids with chromosomes derived from different species. Precisely it is the result of multiplying the chromosome number in an F1 hybrid. *Triticale* is an example of an allopolyploid, having six chromosome sets, allohexaploid, four from wheat (*Triticum turgidum*) and two from rye (*Secale cereale*). *Amphidiploids* are a type of allopolyploids (they are tetraploid, containing the diploid chromosome sets of both parents[57]). Some of the best examples of allopolyploids come from the Brassicas, and the Triangle of U describes the relationships between the three common diploid Brassicas (*B. oleracea, B. rapa,* and *B. nigra*) and three allotetraploids (*B. napus, B. juncea,* and *B. carinata*) derived from hybridization among the diploids.

| Species | Common Name | Family | Hybridization | Confirmed or Putative Hybridization? | Putative Parental/Introgressive species | Polyploid or Homoploid? | Polyploid Chromosome Count | Reference |
|---|---|---|---|---|---|---|---|---|
| Abelmoschus esculentus (L.) Moench | Okra | Malvaceae | Allopolyploid origin | Putative | Uncertain | Polyploid (tetraploid) | usually 2n=4x=130 | Joshi and Ha 1956; Schafleitner al., 2013 |
| Actinidia deliciosa (A. Chev.) C.F.Liang & A.R.Ferguson | Kiwifruit | Actinidiaceae | Allopolyploid origin | Putative | Actinidia chinensis Planch. and Unknown | Polyploid (hexaploid) | 2n=6x=174 | Atkinson et a 1997 |

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| Agave fourcroydes Lem. | Henequen | Asparagaceae | Allopolyploid origin | Confirmed | Uncertain | Polyploid (usu. pentaploid, triploid) | 2n=5x(3x)=150(90) | Robert et al., 2008; Hughe al., 2007 |
| Agave sisalana Perrine | Sisal | Asparagaceae | Allopolyploid origin | Confirmed | Uncertain | Polyploid (usu. pentaploid, hexaploid) | 2n=5x(6x)=150(180) | Robert et al., 2008 |
| Allium ampeloprasum L. | Great headed garlic | Amaryllidaceae | Intraspecific hybrid origin | Putative | Allium ampeloprasum L. | Homoploid | - | Guenaoui et 2013 |
| Allium cepa L. | Common onion | Amaryllidaceae | Interspecific hybrid origin | Putative | Uncertain: Allium vavilovii Popov & Vved., A. galanthum Kar. & Kir. or A. fistulosum L. | Homoploid | - | Gurushidze e 2007 |
| Allium cornutum Clementi | Triploid onion | Amaryllidaceae | Triparental allopolyploid origin | Confirmed | Allium cepa L., A. roylei Stearn, unknown | Polyploid (triploid) | 2n=3x=24 | Fredotovic e 2014 |
| Ananas comosus (L.) Merr is | Pineapple | Bromeliaceae | Interspecific introgression | Putative | Ananas ananassoides (Baker) L.B. Smith | Homoploid | - | Duval et al., |
| Annona x atemoya | Atemoya | Annonaceae | Interspecific hybrid origin | Confirmed | Annona cherimola Mill. and A. squamosa L. | ? | - | Perfectti et a 2004; Jalikop 2010 |
| Arachis hypogaea L. | Peanut | Fabaceae | Allopolyploid origin | Confirmed | Arachis duranensis Krapov. & W.C. Greg. and A. ipaënsis Krapov. & W.C. Greg. | Polypoid (tetraploid) | 2n=4x=40 | Kochert et al 1996; Bertio al., 2011 |
| Armoracia rusticana P.Gaertn., B.Mey. & Scherb. | Horseradish | Brassicaceae | Interspecific hybrid origin | Putative | Uncertain | ? | - | Courter and Rhodes, 196 |
| Artocarpus altilis (Parkinson ex F.A.Zorn) Fosberg | Breadfruit | Moraceae | Interspecific introgression | Putative | Artocarpus mariannensis Trécul | ? | - | Zerega et al., 2005; Jones al., 2013 |
| Avena sativa L. | Oat | Poaceae | Allopolyploid origin | Confirmed | Uncertain | Polyploid (hexaploid) | 2n=6x=42 | Linares et al. 1998; Oliver al., 2013 |
| Brassica carinata A.Braun | Ethiopian mustard | Brassicaceae | Allopolyploid origin | Confirmed | Brassica oleracea L. and B. nigra (L.) K.Koch | Polyploid (tetraploid) | 2n=4x=19 | Arias et al., |
| Brassica juncea (L.) Czern. | Indian mustard | Brassicaceae | Allopolyploid origin | Confirmed | Brassica nigra (L.) K.Koch and B. rapa L. | Polyploid (tetraploid) | 2n=4x=18 | Arias et al., |
| Brassica napus L. | Rapeseed, Rutabega | Brassicaceae | Allopolyploid origin | Confirmed | Brassica rapa L. and B. oleracea L. | Polyploid (tetraploid) | 2n=4x=19 | Arias et al., |
| Cajanus cajan (L.) Millsp. | Pigeon Pea | Fabaceae | Intraspecific introgression, interspecific introgression | Putative | Wild Cajanus cajan and other species | Homoploid | - | Kassa et al., |
| Cannabis sativa L. | Hemp | Cannabaceae | Intraspecific introgression | Putative | Cannabis sativa L. 'Indica' and 'Sativa' types | Homoploid | - | de Meijer an van Soest, 1 |
| Carica pentagona Heilborn | Babaco | Caricaceae | Interspecific hybrid origin | Confirmed | Uncertain (Carica stipulata V.M.Badillo, Vasconcellea pubescens A.DC., Vasconcellea weberbaueri (Harms) V.M. Badillo) | ? | - | Van Droogenbroe et al., 2002; Droogenbroe et al., 2006 |
| Carya illinoinensis (Wangenh.) K.Koch | Pecan | Juglandaceae | Interspecific hybrid origin | Putative | Uncertain | ? | - | Grauke et al. 2011 |
| Castanea dentata (Marshall) Borkh | Chestnut | Fagaceae | Interspecific introgression | Confirmed | Castanea pumila (L.) Mill. | Homoploid | - | Li and Dane, 2013 |
| Castanea | Chestnut | Fagaceae | Intraspecific | Confirmed | Castanea sativa Eurosiberian and | Homoploid | - | Villani et al., 1999; Mattio |

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| Castanea sativa Mill. | Chestnut | Fagaceae | Intraspecific introgression | Confirmed | Eurosiberian and Mediterranean populations | Homoploid | - | 1999; Mattio al., 2013 |
| Chenopodium quinoa Willd. | Quinoa | Chenopodiaceae | Allopolyploid origin | Putative | Uncertain | Polyploid (tetraploid) | - | Heiser, 1974 Ward, 2000; Maughan et a 2004 |
| Cicer arietinum L. | Chickpea (pea-shaped) | Fabaceae | Intraspecific hybrid origin | Putative | Cicer arietinum L. Desi and Kabuli Germplasm | ? | - | Upadhyaya e 2008; Kenen al., 2011 |
| Cichorium intybus L. | Radicchio | Asteraceae | Interspecific introgression | Confirmed | Wild Cichorium intybus L. | Homoploid | - | Kiaer et al., 2 |
| Citrus aurantiifolia (Christm.) Swingle | Key lime | Rutaceae | Interspecific hybrid origin | Confirmed | Citrus medica L. and C. subg. Papeda | ? | - | Ollitrault and Navarro, 201 Penjor et al., 2014; Nicolo al., 2000; Mo 2001 |
| Citrus aurantium L. | Sour oranges | Rutaceae | Interspecific hybrid origin | Confirmed | Citrus maxima (Burm.) and C. reticulata Blanco | ? | - | Wu et al., 20 Moore, 2001 |
| Citrus clementina hort. | Clementine | Rutaceae | Interspecific hybrid origin | Confirmed | Citrus sinensis (L.) Osbeck and C. reticulata Blanco | ? | - | Wu et al., 20 |
| Citrus limon (L.) Osbeck | Lemon, lime | Rutaceae | Interspecific hybrid origin | Confirmed | Citrus medica L., C. aurantiifolia (Christm.) Swingle, and uncertain | ? | - | Nicolosi et a 2000; Moore 2001 |
| Citrus paradisi Macfad. | Grapefruit | Rutaceae | Interspecific hybrid origin | Confirmed | Citrus sinensis (L.) Osbeck and C. maxima (Burm.) | ? | - | Wu et al., 20 Moore, 2001 |
| Citrus reticulata Blanco | Mandarin | Rutaceae | Interspecific introgression | Confirmed | Citrus maxima (Burm.) | ? | - | Wu et al., 20 |
| Citrus sinensis (L.) Osbeck | Sweet orange (blood, common) | Rutaceae | Interspecific hybrid origin | Confirmed | Uncertain | ? | - | Wu et al., 20 Moore, 2001 |
| Cocos nucifera L. | Coconut | Arecaceae | Intraspecific introgression | Confirmed | Cocos nucifera L. Indo-Atlantic and Pacific lineages | Homoploid | - | Gunn et al., 2 |
| Coffea arabica L. | Coffee | Rubiaceae | Allopolyploid origin | Confirmed | Coffea eugenioides S.Moore and C. canephora Pierre ex A.Froehner | Polyploid (tetraploid) | 2n=4x=44 | Lashermes e 1999 |
| Corylus avellana L. | Hazelnut | Betulaceae | Intraspecific introgression | Confirmed | Wild Corylus avellana L. in Southern Europe | Homoploid | - | Campa et al. 2011; Bocca al., 2013 |
| Cucurbita pepo L. | Winter Squash, Pumpkin | Cucurbitaceae | Intraspecific introgression | Putative | Cucurbita pepo var. texana (Scheele) D.S.Decker | Hompoloid | - | Kirkpatrick a Wilson, 1988 |
| Daucus carota subsp. sativus (Hoffm.) Arcang. | Carrot | Apiaceae | Intraspecific introgression | Confirmed | Daucus carota L. subsp. carota | Homoploid | - | Iorizzo et al. 2013; Simon 2000 |
| Dioscorea L. spp. | Yam | Dioscoreaceae | Interspecific hybrid origin, introgression | Putative | Uncertain | Variable | - | Terauchi et a 1992; Dansi al., 1999; Bhattacharje al., 2011; Mignouna et 2002 |
| Diospyros kaki L.f. | Persimmon | Ebenaceae | Allopolyploid origin | Putative | Uncertain | Polyploid (hexaploid) | Yonemori et al., 2008 | |
| Ficus carica L. | Fig | Moraceae | Interspecific introgression | Putative | Uncertain | ? | - Aradhya et al., 2010 | |
| Fragaria ananassa (Duchesne ex Weston) Duchesne ex Rozier | Strawberries | Rosaceae | Interspecific hybrid origin | Confirmed | Fragaria virginiana Mill. (octoploid), F. chiloensis (L.) Mill. (octoploid) | Homoploid relative to parentals (octoploid) | 2n=8x=56 Evans, 1977; Hirakawa et al., 2014 | Uncertain wh species form the octoploid progenitors |

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| Rozier | | | | | | | | |
| Garcinia mangostana L. | Mangosteen | Clusiaceae | Allopolyploid origin | Putative | Garcinia celebica L. and G. malaccensis Hook.f. ex T.Anderson | Polyploid (tetraploid) | ? Richards, 1990 | Recent work shows this m not be of hyb origin (Nazre 2014) |
| Gossypium hirsutum L. | Upland Cotton | Malvaceae | hybrid origin | Confirmed | Uncertain, referred to as 'A' and 'D' | Polyploid (formed <1MYA) | 2n =4x=52 Wendel and Cronn 2003 | Polyploidiza likely led to agronomical significant tr (Applequist e 2001) |
| Hibiscus sabdariffa L. | Roselle | Malvaceae | Allopolyploid origin | Putative | Uncertain | Polyploid (tetraploid) | 2n=4x=72 Menzel and Wilson, 1966; Satya et al., 2013 | |
| Hordeum vulgare L. | Barley | Poaceae | Introgression | Confirmed | Hordeum spontaneum K.Koch | Homoploid | - Badr et al., 2000; Dai et al., 2012 | |
| Humulus lupulus L. | Hops | Cannabaceae | Intraspecific introgression | Confirmed | Humulus lupulus L. North American and European Germplasm | Homoploid | - Reeves and Richards, 2011; Stajner et al., 2008; Seefelder et al., 2000 | |
| Ipomoea batatas (L.) Lam. | Sweet Potato | Convolvulaceae | Intraspecific introgression; Interspecific introgression? | Putative | Ipomoea batatas (L.) Lam. Central American and South American Germplasm | Homoploid relative to parentals | Roullier et al., 2013 | |
| Juglans regia L. | Walnut | Juglandaceae | Interspecific hybridization | Confirmed | Juglans sigillata Dode | Homoploid | - Gunn et al., 2010 | |
| Lactuca sativa L. | Lettuce | Asteraceae | Intraspecific hybrid origin | Putative | Lactuca serriola L. and other L. spp. | Homoploid | - de Vries, 1997 | |
| Lagenaria siceraria (Molina) Standl. | Bottle Gourd | Cucurbitaceae | Intraspecific introgression | Confirmed | Lagenaria siceraria (Molina) Standl. African/American and Asian Germplasm | Homoploid | - Clarke et al., 2006 | |
| Lens culinaris Medik. ssp. culinaris | Lentil | Fabaceae | Intraspecific introgression | Putative | Wild lentil, Lens culinaris subsp. orientalis (Boiss.) Ponert | Homoploid | - Erskine et al., 2011 | |
| Macadamia integrifolia Maiden & Betche | Macadamia | Proteaceae | Interspecific hybrid origin, interspecific introgression | Confirmed | Macadamia tetraphylla L.A.S.Johnson, and other M. spp. | Homoploid | - Hardner et al., 2009; Steiger et al., 2003; Aradhya et al., 1998 | |
| Malus domestica Borkh. | Apple | Rosaceae | Interspecific hybrid origin | Confirmed | Malus sieversii (Ledeb.) M.Roem., M. sylvestris (L.) Mill., and possibly others | Homoploid | - Cornille et al., 2012 | |
| Mentha piperita L. | Peppermint | Lamiaceae | Allopolyploid origin | Confirmed | Mentha aquatica L. and M. spicata L. | Polyploid (12-ploid) | 2n=12x=66 or 72 Harley and Brighton, 1977; Gobert et al., 2002 | |
| Musa paradisiaca L. | Banana | Musaceae | Allopolyploid origin | Confirmed | Musa acuminata Colla, M. balbisiana Colla | Polyploid (usually triploid) | 2n=3x=33 Simmonds and Shepherd, 1955; Heslop-Harrison and Schwarzacher, 2007; De Langhe et al., 2010 | |
| Nicotiana tabacum L. | Tobacco | Solanaceae | Allopolyploid origin | Confirmed | Uncertain (Nicotiana sylvestris Speg. & S. Comes and N. tomentosiformis Goodsp.) | Polyploid (tetraploid) | 2n=4x=48 Kenton et al., 1993; Murad et al., 2002 | |
| Olea europaea L. | Olive | Oleaceae | Intraspecific introgression | Putative | Wild Olea europaea L., Eastern and Western Germplasm | Homoploid | - Kaniewski et al., 2012; Besnard et al., 2013; Breton et al., 2006; Rubio de Casas et al., 2006; Besnard et al., 2007; Besnard et al., 2000 | |
| Opuntia L. spp. | Opuntia | Cactaceae | Interspecific hybrid origin, Allopolyploid origin | Putative | Including Opuntia ficus-indica (L.) Mill. | Polyploid, homoploid | - Hughes et al., 2007; Griffith, 2004 | |
| Oryza sativa L. | Rice | Poaceae | Intraspecific introgression, interspecific introgression | Putative | Oryza sativa L. 'Japonica' and 'Indica' Germplasm, Oryza rufipogon Griff. | Homoploid | - Caicedo et al., 2007; Gao and Innan, 2008 | |
| Oxalis tuberosa Molina | Oca | Oxalidaceae | Allopolyploid origin | Putative | Uncertain | Polyploid (octaploid) | 2n=8x=64 Emswhiller and Doyle, 2002; Emshwiller 2002; Emswhiller et al., 2009 | |

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| Pennisetum glaucum (L.) R.Br. | Pearl Millet | Poaceae | Intraspecific introgression | Confirmed | Wild Pennisetum glaucum (L.) R.Br. | Homoploid | - Oumar et al., 2008 | |
| Persea americana Mill. | Avocado (Hass and other cultivars) | Lauraceae | Intraspecific introgression | Confirmed | Persea americana Mill. 'Guatamalensis', 'Drymifolia', and' Americana' | Homoploid | - Chen et al., 2008; Davis et al., 1998; Ashworth and Clegg, 2003; Douhan et al., 2011 | |
| Phoenix dactylifera L. | Date palm | Arecaceae | Interspecific hybrid origin | Putative | Uncertain | Homoploid | - El Hadrami et al., 2011; Bennaceur et al., 1991 | |
| Piper methysticum G.Forst. | Kava | Piperaceae | Allopolyploid origin | Putative | Piper wichmannii C. DC. and P. gibbiflorum C.DC. | Polyploid (decaploid) | 2n=10x=130 Singh, 2004; Lebot et al., 1991 | |
| Pistacia vera L. | Pistachio | Anacardiaceae | Interspecific introgression | Putative | Pistacia atlantica Desf., P. chinensis subsp. integerrima (J. L. Stewart ex Brandis) Rech. f. | Homoploid | - Kafkas et al., 2001 | |
| Pisum abyssinicum A.Braun | Pea | Fabaceae | Interspecific hybrid origin | Confirmed | Uncertain (Pisum fulvum Sibth. & Sm. and other P. spp.) | Homoploid | - Vershinin et al., 2003 | |
| Pisum sativum L. | Pea | Fabaceae | Interspecific hybrid origin | Confirmed | Uncertain (Pisum sativum subsp. elatius (M.Bieb.) Asch. & Graebn. and other P. spp.) | Homoploid | - Vershinin et al., 2003 | |
| Plinia cauliflora (Mart.) Kausel | Jaboticaba | Myrtaceae | Intraspecific hybridization, interspecific hybridization | Putative | Plinia 'Jaboticaba' and 'Cauliflora' Germplasm; P. peruviana (Poir.) Govaerts | Homoploid | - Balerdi et al., 2006 | |
| Prunus cerasus L. | Cherry | Rosaceae | Allopolyploid origin | Confirmed | Prunus avium (L.) L. and P. fruticosa Pall. | Polyploid (tetraploid) | 2n=4x=32 Tavaud et al., 2004; Olden and Nybom, 1968 | |
| Prunus domestica L. | Plum | Rosaceae | Allopolyploid origin | Confirmed | Uncertain (P. cerasifera Ehrh. and P. spinosa L.) | Polyploid (hexaploid) | 2n=6x=48 Zohary, 1992; Hartmann and Neumuller, 2009 | Japanese Plu also of hybri origin (see Hartmann an Neumuller 2 Also hybridi with other cultivated Pr spp. |
| Prunus dulcis (Mill.) D.A.Webb | Almond | Rosaceae | Interspecific introgression | Confirmed | Prunus orientalis (Mill.) Koehne and other P. spp. | Homoploid | - Delplancke et al., 2012; Delplancke et al., 2013 | |
| Pyrus L. species | Pear | Rosaceae | Interspecific hybrid origin | Confirmed | Many species | Homoploid | - Silva et al., 2014 | Also introgression with semidomesti populations ( Iketani et al. 2009) |
| Raphanus raphanistrum subsp. sativus (L.) Domin | Radish | Brassicaceae | Intraspecific introgression | Confirmed | Raphanus raphanistrum L. subsp. raphanistrum | Homoploid | - Ridley et al., 2008 | |
| Rheum L. cultivated species | Rhubarb | Polygonaceae | Interspecific hybrid origin | Putative | Unclear | Homoploid relative to parentals (tetraploid) | - Foust and Marshall, 1991; Kuhl and Deboer, 2008 | Hybrids incl Rheum rhaponticum R. rhabarbar L., R. palmat L. |
| Rubus L. spp. | Red raspberry, Blackberry, Tayberry, Boysenberry, etc. | Rosaceae | Allopolyploid origin, interspecific hybridization | Confirmed | Many | Polyploid | - Alice et al., 2014; Alice and Campbell, 1999 | |
| Saccharum spp. | Sugarcane | Poaceae | Allopolyploid origin | Confirmed | Saccharum officinarum L. and S. spontaneum L. | Polyploid | Variable, 2n=10-13x=100-130 Grivet et al., 1995; D'Hont et al., 1996 | |
| Secale cereale L. | Rye | Poaceae | Interspecific hybrid origin | Confirmed | Uncertain (Secale montanum Guss., S. vavilovii Grossh.) | Homoploid | - Bartos et al., 2008; Korzun et al., 2001; Hillman, 1978; Tang et al., 2011; Salamini et al., 2002 | |
| Sechium edule (Jacq.) Sw. | Chayote | Cucurbitaceae | Interspecific introgression | Putative | Sechium compositum (Donn. Sm.) C. Jeffrey | Homoploid | - Newstrom, 1991 | |
| Setaria italica | Foxtail | | Interspecific | | Setaria viridis (L.) | | - Till-Bottraud et al. | |

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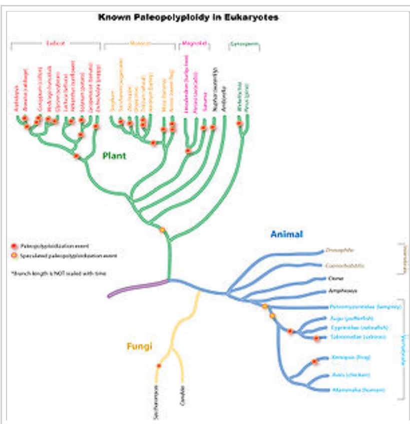
| Setaria italica (L.) P.Beauv. | Foxtail millet | Poaceae | Interspecific introgression | Confirmed | Setaria viridis (L.) P.Beauv. | Homoploid | - Till-Bottraud et al., 1992 | |
| Solanum L. spp. Section Petota | Potato | Solanaceae | Interspecific hybrid origin, allopolyploid origin, interspecific introgression | Confirmed | Including Solanum tuberosum L., S. ajanhuiri Juz. & Bukasov, S. curtilobum Juz. & Bukasov, S. juzepczukii Bukasov | Homoploid and Polyploid | - Rodriguez et al., 2010 | |
| Solanum lycopersicum L. | Tomato | Solanaceae | Intraspecific introgression, interspecific introgression | Confirmed | Solanum lycopersicum var. cerasiforme (Dunal) D.M. Spooner, G.J. Anderson & R.K. Jansen and S. pimpinellifolium L. | Homoploid | - Blanca et al., 2012; Causse et al., 2013; Rick, 1950 | |
| Solanum melongena L. | Eggplant | Solanaceae | Interspecific hybrid origin, interspecific introgression, intraspecific introgression | Confirmed | Solanum undatum Lam. and others; wild S. melongena L. (=S. insanum L.) | Homoploid | - Knapp et al., 2013; Meyer et al., 2012 | Hybrid origi not confirme but introgres is well documented |
| Solanum muricatum Aiton | Pepino dulce | Solanaceae | Interspecific hybrid origin, interspecific introgression | Likely | Solanum species in Series Caripensia | Homoploid | - Blanca et al., 2007 | Polyphyletic origin and extensive, ongoing introgression with wild sp |
| Spondias purpurea L. | Jocote | Anacardiaceae | Interspecific introgression | Putative | Spondias mombin L. | Homoploid | - Miller and Schaal, 2005 | |
| Theobroma cacao L. | Cacao (Trinitario-type) | Malvaceae | Intraspecific hybrid origin | Confirmed | Theobroma cacao L. 'Forastero' and 'Criollo' Germplasm | Homoploid | - Yang et al., 2013 | |
| Triticum aestivum L. | Bread Wheat, Spelt | Poaceae | Allopolyploid origin | Confirmed | Triticum turgidum L. (tetraploid) with Aegilops tauschii Coss. | Polyploid (hexaploid) | 2n=6x=42 Matsuoka, 2011; Dvorak, 2012 | |
| Triticum turgidum L. | Emmer Wheat, Durum Wheat | Poaceae | Allopolyploid origin | Confirmed | Triticum urartu Thumanjan ex Gandilyan and Aegilops speltoides Tausch | Polyploid (tetraploid) | 2n=4x=28 Dvorak et al., 2012; Matsuoka, 2011; Yamane and Kawahara, 2005 | |
| Vaccinium corymbosum L. | Highbush Blueberry | Ericaceae | Interspecific hybrid origin, interspecific introgression | Putative | Vaccinium tenellum Aiton, V. darrowii Camp, (V. virgatum Aiton, V. angustifolium Aiton) | Uncertain | - Vander Kloet, 1980; Bruederle et al., 1994; Lyrene et al., 2003; Boches et al., 2006 | Possible hyb origin during Plesitocene |
| Vanilla tahitensis J.W. Moore | Tahitian vanilla | Orchidaceae | Allopolyploid origin | Confirmed | Vanilla planifolia Jacks. ex Andrews and V. odorata C.Presl | Polyploid | Variable, 2n=2x(4x)=32(64) Lubinsky et al., 2008 | |
| Vitis rotundifolia Michx. | Grape | Vitaceae | Interspecific introgression | Confirmed | ManyVitis spp. | Homoploid | 2n=2x=38 Reisch et al., 2012; This et al., 2006 | |
| Zea mays L. | Maize | Poaeceae | Intraspecific introgression | Confirmed | Wild Zea mays L. (teosinte, =subsp. parviglumis Iltis & Doebley) | Homoploid | - Van Heerwaarden et al., 2011; Hufford et al., 2013 | |

[58]

## Paleopolyploidy

Ancient genome duplications probably occurred in the evolutionary history of all life. Duplication events that occurred long ago in the history of various evolutionary lineages can be difficult to detect because of subsequent diploidization (such that a polyploid starts to behave cytogenetically as a diploid over time) as mutations and gene translations gradually make one copy of each chromosome unlike the other copy. Over time, it is also common for duplicated copies of genes to accumulate mutations and become inactive pseudogenes.[59]



This phylogenetic tree shows the relationship between the best-documented instances of paleopolyploidy in eukaryotes.

In many cases, these events can be inferred only through comparing sequenced genomes. Examples of unexpected but recently confirmed ancient genome duplications include baker's yeast (*Saccharomyces cerevisiae*), mustard weed/thale cress (*Arabidopsis thaliana*), rice (*Oryza sativa*), and an early evolutionary ancestor of the vertebrates (which includes the human lineage) and another near the origin of the teleost fishes. Angiosperms (flowering plants) have paleopolyploidy in their ancestry. All eukaryotes probably have experienced a polyploidy event at some point in their evolutionary history.

## Karyotype

A karyotype is the characteristic chromosome complement of a eukaryote species.[60][61] The preparation and study of karyotypes is part of cytology and, more specifically, cytogenetics.

Although the replication and transcription of DNA is highly standardized in eukaryotes, the same cannot be said for their karotypes, which are highly variable between species in chromosome number and in detailed organization despite being constructed out of the same macromolecules. In some cases, there is even significant variation within species. This variation provides the basis for a range of studies in what might be called evolutionary cytology.

#### Paralogous

The term is used to describe the relationship between duplicated genes or portions of chromosomes that derived from a common ancestral DNA. Paralogous segments of DNA may arise spontaneously by errors during DNA replication, copy and paste transposons, or whole genome duplications.

#### Homologous

The term is used to describe the relationship of similar chromosomes that pair at mitosis and meiosis. In a diploid, one homolog is derived from the male parent (sperm) and one is derived from the female parent (egg). During meiosis and gametogenesis, homologous chromosomes pair and exchange genetic material by recombination, leading to the production of sperm or eggs with chromosome haplotypes containing novel genetic variation.

#### Homoeologous

The term *homoeologous*, also spelled *homeologous*, is used to describe the relationship of similar chromosomes or parts of chromosomes brought together following inter-species hybridization and allopolyploidization, and whose relationship was completely homologous in an ancestral species. In allopolyploids, the homologous chromosomes within each parental sub-genome should pair faithfully during meiosis, leading to disomic inheritance; however in some allopolyploids, the homoeologous chromosomes of the parental genomes may be nearly as similar to one another as the homologous chromosomes, leading to tetrasomic inheritance (four chromosomes pairing at meiosis), intergenomic recombination, and reduced fertility.

##### Example of homoeologous chromosomes

Durum wheat is the result of the inter-species hybridization of two diploid grass species *Triticum urartu* and *Aegilops speltoides*. Both diploid ancestors had two sets of 7 chromosomes, which were similar in terms of size and genes contained on them. Durum wheat contains two sets of chromosomes derived from *Triticum urartu* and two sets of chromosomes derived from *Aegilops speltoides*. Each chromosome pair derived from the *Triticum urartu* parent is **homoeologous** to the opposite chromosome pair derived from the *Aegilops speltoides* parent, though each chromosome pair unto itself is **homologous**.

## Polyploidization and speciation

Polyploidization is a mechanism of sympatric speciation because polyploids are usually unable to interbreed with their diploid ancestors. An example is the plant *Mimulus peregrinus*. Sequencing confirmed that this species originated from *M. x robertsii*, a sterile triploid hybrid between *M. guttatus* and *M. luteus,* both of which have been introduced and naturalised in the United Kingdom. New populations of *M. peregrinus* arose on the Scottish mainland and the Orkney Islands via genome duplication from local populations of *M. x robertsii*.[62]

## See also

- Polyploid complex
- Polysomy
- Sympatry

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## External links

- Polyploidy on Kimball's Biology Pages (http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/P/Polyploidy.html)
- The polyploidy portal (http://www.polyploidy.org/) a community-editable project with information, research, education, and a bibliography about polyploidy.